Ruo-Yu Guan

Curriculum vitae · November 2026
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Research Interests

- Space-based gravitational-wave astrophysics; Galactic compact binaries, particularly double white dwarfs.
- Unresolved Galactic-binary gravitational-wave foregrounds and angular anisotropy; residual foreground statistics and non-Gaussianity.
- Stochastic gravitational-wave backgrounds and statistical inference.
- Potential applications to Galactic dynamics, dark matter, and unusual astrophysical environments.

Education

Huazhong University of Science and Technology — Wuhan, China
School of Physics; National Gravitation Laboratory — *Sep. 2025 – Jun. 2029 (expected)*
Ph.D. in Physics (Theoretical Physics) — *Supervisor: Prof. Yan Wang*

- Doctoral coursework completed; thesis proposal successfully defended.
- Ranked 2nd of 138 Ph.D. students based on course grades.

The University of Hong Kong — Hong Kong SAR, China
M.Sc. in Physics, Department of Physics, Faculty of Science — *Degree conferred: Nov. 2024*

- Capstone project: *Identifying Sources in Cygnus OB2 Using Multiwavelength Observations*, supervised by Assoc. Prof. Stephen Chi-Yung Ng.

Jilin University — Changchun, China
B.Sc. — *Degree conferred: Jun. 2022*

- Graduation thesis: *Study on the Prompt Emission of Gamma-ray Bursts and its Polarization*; research mentor: Prof. Mi-Xiang Lan.

Research Positions

Huazhong University of Science and Technology — Wuhan, China
Research Assistant (contract), Department of Astronomy — *Nov. 2024 – Jun. 2025*

- Gamma-ray burst research with Prof. Yuan-Chuan Zou.

Publications

1. **Ruo-Yu Guan** and Yan Wang, Residual Galactic binary foreground in LISA stochastic gravitational-wave background inference: Source power concentration and spectral degeneracy, *Physical Review D*, accepted for publication (2026).
   DOI: 10.1103/4779-jw6l · arXiv:2607.25349 · Code & Data (Zenodo)
2. **Ruo-Yu Guan**, Fei-Fei Wang, and Yuan-Chuan Zou, Hurst index of gamma-ray burst light curves and its statistical study, *Journal of High Energy Astrophysics* **51**, 100559 (2026).
   DOI: 10.1016/j.jheap.2026.100559
3. **Ruo-Yu Guan** and Mi-Xiang Lan, Interpreting time-integrated polarization data of gamma-ray burst prompt emission, *Astronomy & Astrophysics* **670**, A160 (2023).
   DOI: 10.1051/0004-6361/202243805

Research Experience

Angular Anisotropy of the Unresolved Galactic-Binary Foreground — Ongoing
National Gravitation Laboratory and School of Physics, Huazhong University of Science and Technology — *Wuhan, China*
Research with Prof. Yan Wang

- Studying the angular anisotropy of the unresolved Galactic-binary gravitational-wave foreground, with an emphasis on double white dwarf populations.
- Investigating how the spatial distribution of Galactic sources shapes the sky-dependent foreground for space-based detectors.

Residual Galactic Binary Foreground and SGWB Inference — Completed
National Gravitation Laboratory and School of Physics, Huazhong University of Science and Technology — *Wuhan, China*
Research with Prof. Yan Wang

- Estimated the catalog-level residual foreground using LISA Data Challenge 2A injections and the Erebor injection–recovery match.
- Quantified source-discreteness signatures through excess kurtosis, including their dilution by instrumental noise, and analyzed spectral degeneracy with an isotropic stochastic gravitational-wave background.
- First-author paper accepted for publication in *Physical Review D* (2026); supporting analysis code and processed numerical results released through Zenodo.

Time-Series and Correlation Analysis of Gamma-Ray Burst Light Curves Oct. 2024 – Jan. 2026
Department of Astronomy, Huazhong University of Science and Technology *Wuhan, China*
Research with Prof. Yuan-Chuan Zou

- Analyzed 163 long BATSE bursts using detrended fluctuation analysis and Hurst-index estimation; performed statistical correlation studies in Python.
- Resulted in a first-author refereed article in the *Journal of High Energy Astrophysics*.

Identifying Sources in Cygnus OB2 Using Multiwavelength Observations Sep. 2023 – Jun. 2024
Department of Physics, Faculty of Science, The University of Hong Kong *Hong Kong SAR, China*
M.Sc. capstone project supervised by Assoc. Prof. Stephen Chi-Yung Ng

- Reduced and merged 18 Chandra observations and assembled a catalog of detected X-ray sources.
- Cross-matched X-ray sources with Gaia counterparts to distinguish association members from foreground and background objects.

Gamma-Ray Burst Prompt Emission and Polarization 2022 – Feb. 2023
Center for Theoretical Physics and College of Physics, Jilin University *Changchun, China*
Research with Prof. Mi-Xiang Lan, originating in the B.Sc. thesis project

- Computed time-integrated polarization degrees for 37 gamma-ray bursts under a synchrotron-emission framework and compared the predictions with observations.
- Resulted in a first-author refereed article in *Astronomy & Astrophysics*.

Conference Presentations

Shanghai International Symposium on Relativistic Astrophysics Dec. 2026
Scheduled poster presentation: "Residual Galactic binary foreground in LISA stochastic gravitational-wave background inference"
Shanghai, China · Presentation details

2026 Mid-Year Workshop of the National Key R&D Program Project on Space Gravitational-Wave Detection Signal Simulation, Processing, and Verification Jul. 2026
Oral presentation: "Galactic Binary Residuals in LISA SGWB Amplitude Inference: Source Power Concentration and Spectral Degeneracy"
Yangzhou, China · Presentation and slides

2024 PKU International PhD Student Forum on the Frontiers of Modern Astronomy Dec. 2024
Poster presentation: "Time-integrated Polarizations of Gamma-ray Burst Prompt Phase"
Beijing, China · Presentation and poster

Academic Memberships

- **Laser Interferometer Space Antenna (LISA) Consortium** — Core Member, Astrophysics Working Group, Mar. 2026 – Present.
- **Laboratory for Space Research, The University of Hong Kong** — Member, Oct. 2023 – Nov. 2024.

Skills

- **Research Methods:** Statistical analysis and inference (Fisher-matrix analysis, correlation analysis); time-series analysis; binary population synthesis (COSMIC).
- **Programming & Tools:** Python (NumPy, SciPy, pandas, Matplotlib); MATLAB, Mathematica, R (familiar); Git, Linux, LaTeX.
- **Languages:** Chinese (Mandarin: native; Cantonese: conversational); English (English-taught M.Sc. at HKU).